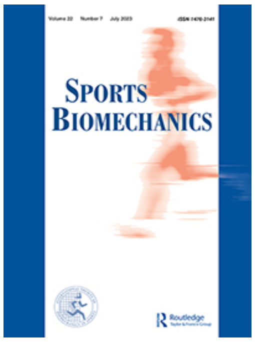

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Cumulative patellofemoral force and stress are lower during faster running compared to slower running in recreational runners

Eoin W. Doyle[a,b], Tim L. A. Doyle[a,b], Jason Bonacci[c], Aaron J. Beach[a] and Joel T. Fuller[a,b]

[a]Faculty of Medicine, Health, and Human Sciences, Macquarie University, Sydney, New South Wales, Australia; [b]Biomechanics, Physical Performance, and Exercise Research Group, Macquarie University, Sydney, New South Wales, Australia; [c]Centre for Sports Research, School of Exercise and Nutrition Sciences, Deakin University, Burwood, Victoria, Australia

ABSTRACT
Management strategies for patellofemoral pain often involve modifying running distance or speed. However, the optimal modification strategy to manage patellofemoral joint (PFJ) force and stress accumulated during running warrants further investigation. This study investigated the effect of running speed on peak and cumulative PFJ force and stress in recreational runners. Twenty recreational runners ran on an instrumented treadmill at four speeds (2.5–4.2 m/s). A musculoskeletal model derived peak and cumulative (per 1 km of continuous running) PFJ force and stress for each speed. Cumulative PFJ force and stress decreased with faster speeds (9.3–33.6% reduction for 3.1–4.2 m/s vs. 2.5 m/s). Peak PFJ force and stress significantly increased with faster speeds (9.3–35.6% increase for 3.1–4.2 m/s vs. 2.5 m/s). The largest cumulative PFJ kinetics reductions occurred when speeds increased from 2.5 to 3.1 m/s (13.7–14.2%). Running at faster speeds increases the magnitude of peak PFJ kinetics but conversely results in less accumulated force over a set distance. Selecting moderate running speeds (~3.1 m/s) with reduced training duration or an interval-based approach may be more effective for managing cumulative PFJ kinetics compared to running at slow speeds.

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Running; speed; kinetics; patellofemoral joint; knee

Introduction

Patellofemoral pain (PFP) is a common musculoskeletal condition with an annual prevalence of approximately 23% in the general population (Smith, Selfe, et al., 2018). It is characterised by an insidious onset of pain localised to the anterior retropatellar or peripatellar region of the knee (Willy et al., 2019). More than 50% of patients with PFP report persistent symptoms after 5–20 years (Lankhorst et al., 2016; Nimon et al., 1998), which can lead to fear avoidant behaviours (Smith, Moffatt, et al., 2018) or restricted physical activity (Rathleff et al., 2016). PFP is typically aggravated by activities that load the patellofemoral joint (PFJ), such as running. PFP is the most commonly reported running-related injury, accounting for up to 17% of injuries (Francis et al., 2019).

CONTACT Eoin W. Doyle eoin.doyle@mq.edu.au
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The aetiology of PFP is considered to be multifactorial (Willy et al., 2019). PFP is associated with biomechanical, psychosocial, and lifestyle factors (Bazett-Jones et al., 2023; Crossley et al., 2016). The specific role of biomechanics in the development of PFJ remains debated (Bazett-Jones et al., 2023). During running, the PFJ is exposed to forces of approximately 4 to 6 times body weight (Willy et al., 2016) resulting in upwards of 320 body weights accumulated per minute of running (Esculier et al., 2020). Excessive PFJ loading is widely thought to contribute to the development and persistence of PFP in most cases (Powers et al., 2012; Willy et al., 2019). It has also been suggested that the excessive PFJ loading associated with PFP relates more to excessive running volume or mileage rather than faster running speeds (Nielsen et al., 2013).

While the traditional biomechanical focus of PFP researchers has been on modifying the kinematics and kinetics of problematic movements, contemporary approaches suggest that decreasing the external training loads acting on an irritated PFJ may be warranted to resolve the problem (Esculier et al., 2020). According to the latest PFP clinical guidelines (Willy et al., 2019) and consensus statement (Collins et al., 2018), optimal management strategies of runners with PFP should target a reduction in the relative force and stress applied to the PFJ. Such strategies may include recommendations to reduce the running distance or speed (Esculier et al., 2020) to lessen the external loads acting on the PFJ. When determining these external loads, it is important to consider not just the peak load per stride but also the accumulated or 'cumulative load' accrued during running. Cumulative load can be viewed as the accumulation of force or stress over the total number of strides or loading cycles for a given distance. The accumulated load is arguably more relevant to PFJ problems compared to peak loads from individual steps, given that greater peak PFJ loads are traditionally observed in other tasks, such as during squatting and jumping activities (Hart et al., 2022).

The influence of running speed on cumulative PFJ kinetics has been explored in a small number of studies. A recent study including high-level runners reported that faster running resulted in higher peak PFJ stresses compared to slow running but did not observe any difference in weighted cumulative PFJ force across speeds (Starbuck et al., 2021). The study examined running speeds greater than 3.3 m/s which are faster than the typical training speeds of recreational runners. The average global marathon running time is estimated to be approximately 4 hours, 32 minutes, or 2.5 m/s (Andersen, 2019), so previous results for PFJ force and stress may not be generalisable to this recreational running population. Recreational runners experience comparatively higher rates of running-related injuries (Buist et al., 2010), so it is important to ensure that biomechanical research is relevant to them (Buist et al., 2010). Another study that included recreational runners reported greater cumulative knee extensor moment impulse during slower running speeds compared to faster speeds, primarily due to the increased number of steps required to complete the same distance (Petersen et al., 2015). This study investigated running speeds ranging from 2.2 to 4.4 m/s, more representative of typical recreational running speeds. However, they did not report on PFJ force and stress outcomes.

PFP load management advice should consider the cumulative PFJ load and not just peak values from individual steps. Modifying running speeds in recreational runners may result in conflicting changes to the peak and cumulative load experienced by runners due to the increased number of steps required to run set training distances at slower speeds. It is still unclear whether the increased number of steps needed to run a given distance results in a similar or reduced cumulative PFJ force and stress across the typical training speeds adopted by recreational runners. This knowledge could provide greater specificity in training guidance around running speeds for set distances and may aid coaches and clinicians in optimally modifying joint loads in runners with PFP. Therefore, this study aims to investigate the influence of speed on peak and cumulative PFJ force and stress per km in recreational runners across speeds (2.5–4.2 m/s) reflective of recreational running. It is hypothesised that peak PFJ force and stress will increase with faster running speeds, but cumulative loads per km will decrease.

Materials and methods

Participants

Twenty (10 male and 10 female) healthy recreational runners were recruited. An a priori sample size calculation conducted in G*Power 3.1 (Universität Kiel, Germany), using an effect size of 0.7 (medium), a power of 0.8 and a critical alpha of 0.05, determined a sample size of at least 19 participants was required. The sample size was in line with previous similar biomechanical studies (Bonacci et al., 2014; Petersen et al., 2015; Starbuck et al., 2021). Participants were recruited through current participant databases and online advertisements in running groups. Participants were included if they were aged between 18 and 50 years and running at least 10 km per week on average over the previous 3 months. Participants were excluded if they reported a history of lower limb surgery, a running-related lower limb injury within the previous 6 weeks, or any neurological, inflammatory, or rheumatoid disease that affected their running gait. All participants completed written informed consent before enrolment. Ethical approval was granted by the Macquarie University Human Research Ethics Committee (Reference No: 52020790819029).

Procedures

Participants attended two laboratory testing sessions. In the first session, participants' height (m), body mass (kg), and running history were collected. Following a self-selected warm-up including at least 5 minutes of treadmill running, participants were fitted with a portable metabolic analysis system (K5 COSMED, Italy) and a heart rate monitor (Garmin HRM Dual, Olathe, KS, the United States). To provide descriptive data on participants' athletic ability, they completed a multi-stage maximal running assessment on a treadmill to determine VO_2 max (ml/kg/min) based on established protocols from the Australian Institute of Sport (Tanner & Gore, 2013) that have demonstrated excellent reliability (CV: 1.3–5.6%) (Hopkins et al., 2001). In the second session, participants performed continuous three-minute treadmill runs at up to four speeds (2.5, 3.1, 3.6 and 4.2 m/s). These speeds were chosen to represent a range of training paces for healthy recreational runners, with consideration for the slower running speeds, and consistent with paces used in previous PFP research (Hart et al., 2022). Adequate recovery time was provided between running trials to ensure heart rate returned to resting levels below 60% of maximum heart rate. A Vicon motion capture system with eight cameras (250 Hz, Oxford Metrics Ltd, Oxford, UK) was utilised to acquire 3D kinematic data of the lower limb while simultaneously collecting synchronised ground reaction force data at 1000 Hz from a force-instrumented treadmill (AMTI Compact Tandem, Massachusetts, US). A University of Western Australia full-body marker set (Chin et al., 2009) comprising 15 tracking and 11 calibration spherical retro-reflective markers (12.7 mm diameter) and marker clusters were placed on the participant bilaterally on select landmarks at the head, torso, pelvis, and legs (Appendix A). Static calibration was performed with participants in the standard anatomically neutral position.

Data analysis

Data collected during biomechanical analysis was processed using NEXUS (v2.14, Vicon, Oxford, UK). Automated labelling algorithms were utilised to label markers, which were manually checked and adjusted for accuracy. Missing marker trajectories (less than 12 frames) were filled using either a Woltring quintic spline method (Woltring, 1985) or a rigid body fill. Data were exported to Visual 3D (v5; C-Motion, Inc., MD, United States), where a nine-segment lower extremity model was built, and 3D kinematic and kinetic variables were determined. Static calibration trials were conducted using calibration markers to define joint centres and track body segments (Cappozzo et al., 1995). Hip joint centres were predicted using equations from Bell et al. (1990). Knee and ankle joint centres were predicted using the midpoint of the medial and lateral femoral condyles and malleoli at the knee and ankle, respectively. Gait events were determined based on vertical GRF thresholds of 50 N (Napier et al., 2021). Kinematic and force plate data were filtered using a low-pass fourth-order Butterworth filter with a matched cut-off frequency of 20 Hz (Mai & Willwacher, 2019) via a fourth-order Butterworth low-pass filter (Robertson & Dowling, 2003). Temporospatial parameters, including step rate and step length, joint angles and moments were determined. The net internal knee extension moment was determined by submitting filtered kinematic and GRF to a conventional Newton-Euler inverse dynamics analysis (Bonacci et al., 2018) and reported in the proximal segment coordinate system. Joint moments were normalised to body mass*height and reported in Nm/kg·m.

PFJ force was calculated using a commonly used biomechanical model (Bonacci et al., 2014; Heino Brechter & Powers, 2002) that considers knee flexion angle, net knee extension moment, and the effective quadriceps moment arm (Table 1). The model follows the recent recommendations from Nunes et al. (2018). and is similar to other

Table 1. Equations.

Number	Equation
1	$CA_{PFJ} = 2.0\ e^{-5}x^4 - 0.0033x^3 + 0.1099x^2 + 3.5273x + 81.058$
2	$L_{eff} = 8.0\ e^{-5}x^3 - 0.013x^2 + 0.28x + 46$
3	$F_q = (M_k + [F_{ham} \times MA_{ham}] + [F_{Achilles} \times MA_{Achilles}])/L_{eff}$
4	$k = (-3.8\ e^{-5}x^2 + 1.47e^{-3}x + 0.462) \div (-6.98\ e^{-7}x^3 + 1.55\ e^{-4}x^2 - 0.0162x + 1)$
5	$JRF_{PFJ} = k \times F_q$
6	$PFJ_{Stress} = JRF_{PFJ} \div CA_{PFJ}$

Abbreviations: CA_{PFJ}, contact area of the patellofemoral joint in mm^2; F_q, quadriceps force; F_{ham}, hamstring force; $F_{gastroc}$, gastrocnemius force; MA_{ham}, estimated hamstring moment arm at the hip; $MA_{Achilles}$, Achilles tendon moment arm at the ankle; JRF_{PFJ}, Patellofemoral joint reaction force; k, coefficient; L_{eff}, effective lever arm; M_k, knee extensor moment; PFJ_{Stress}, patellofemoral joint stress; x, knee flexion angle.

studies that calculated PFJ force (Bonacci et al., 2014; Starbuck et al., 2021). Quadriceps force was adjusted to account for co-contraction of the knee flexors and extensors using the methods described by DeVita and Hortobagyi (2001), where hamstring and gastrocnemius forces were multiplied by their estimated moment arms at the knee joint as a function of knee angle and added to the knee extensor moment. The quadriceps effective lever arm was determined by fitting a non-linear equation using data obtained from van Eijden et al. (1986). PFJ loading rate was calculated as the maximum slope of the PFJ force curve between initial contact and peak PFJ force. Cumulative PFJ force and stress were estimated from the product of respective total impulse per step (sum of positive and negative) and the number of strides per 1 km of continuous running (500 m/step length) (Willy et al., 2016). Impact variables, including peak GRF and instantaneous vertical loading rate (IVLR), were calculated to assist with interpretation based on methods by Futrell et al. (2018). Briefly, a point of interest (i.e., impact peak) was determined at the first instance above 75% of the participant's body weight with a vertical GRF of less than 15 BW/s. IVLR was then defined as the peak vertical loading rate between 20% and 100% of the GRF curve between the footstrike and the point of interest.

To ensure data consistency, all variables were averaged across 24 seconds during the middle third of the 3-minute trial for each participant to ensure at least 30 consecutive steps were recorded (Oliveira & Pirscoveanu, 2021). Only the right limb was analysed for all participants.

Statistical analyses

Statistical analyses were conducted using Jamovi (version 1.6, Sydney, Australia). Linear mixed models were used to examine the relationships between the dependent variables of peak and cumulative PFJ force and stress and the independent variable of running speed. Models included speed as a fixed effect and participant as a random effect. Secondary biomechanical outcomes, including spatiotemporal (e.g., step rate) and kinematic variables (e.g., peak knee flexion during stance), were investigated using the same modelling approach. Assumptions of the models were assessed by examining residual plots to determine that residuals were approximately normal distribution with homogeneity of variances. Post-hoc t-tests were utilised to determine specific pairwise differences between running speeds. Statistical significance was set at 0.05. Descriptive data were presented as group means (SD) and percentage differences. Effect sizes (ES) were calculated using Cohen's *d*, with 0.2, 0.5, and 0.8 representing small, medium, and large ES, respectively (Cohen, 2013).

Results

Participant characteristics are displayed in Table 2. All participants completed runs at the 2.5, 3.1, and 3.6 m/s speeds, but only 18/20 participants completed runs at the 4.2 m/s speed. Two participants could not complete the 4.2 m/s speed due to physical exhaustion and were excluded from analysis at this speed. Most runners (15/20) primarily adopted a rearfoot footstrike landing pattern, while five runners adopted a mid-to-forefoot footstrike landing pattern across all running speeds.

Table 2. Participant characteristics (mean ± standard deviation).

Characteristic	Male ($n = 10$)	Female ($n = 10$)	Total ($n = 20$)
Age, years	34.0 ± 7.4	34.1 ± 11.9	34.0 ± 9.7
Height, m	1.74 ± 0.05	1.66 ± 0.04	1.70 ± 0.06
Mass, kg	70.4 ± 7.2	59.3 ± 5.3	64.8 ± 8.4
Weekly distance, km	57.0 ± 23.6	50.0 ± 22.9	53.3 ± 22.9
10 km race time, min	39.7 ± 3.8	46.4 ± 3.9	43.2 ± 5.1
VO_{2MAX}, ml/kg/min	65.8 ± 8.6	57.3 ± 5.2	61.4 ± 8.1

Abbreviations: VO_{2Max}, Maximal rate of oxygen used per kilogram of body mass per minute.

Peak PFJ force ($p < 0.001$) and stress ($p < 0.001$) both significantly increased with running speed (Figure 1(a)). Post hoc tests showed that the largest increases in PFJ force and stress occurred between the slowest running speed (2.5 m/s) and the two fastest speeds (3.6 and 4.2 m/s), with a 22.0–35.4% increase in PFJ force (ES 0.90–1.48) and

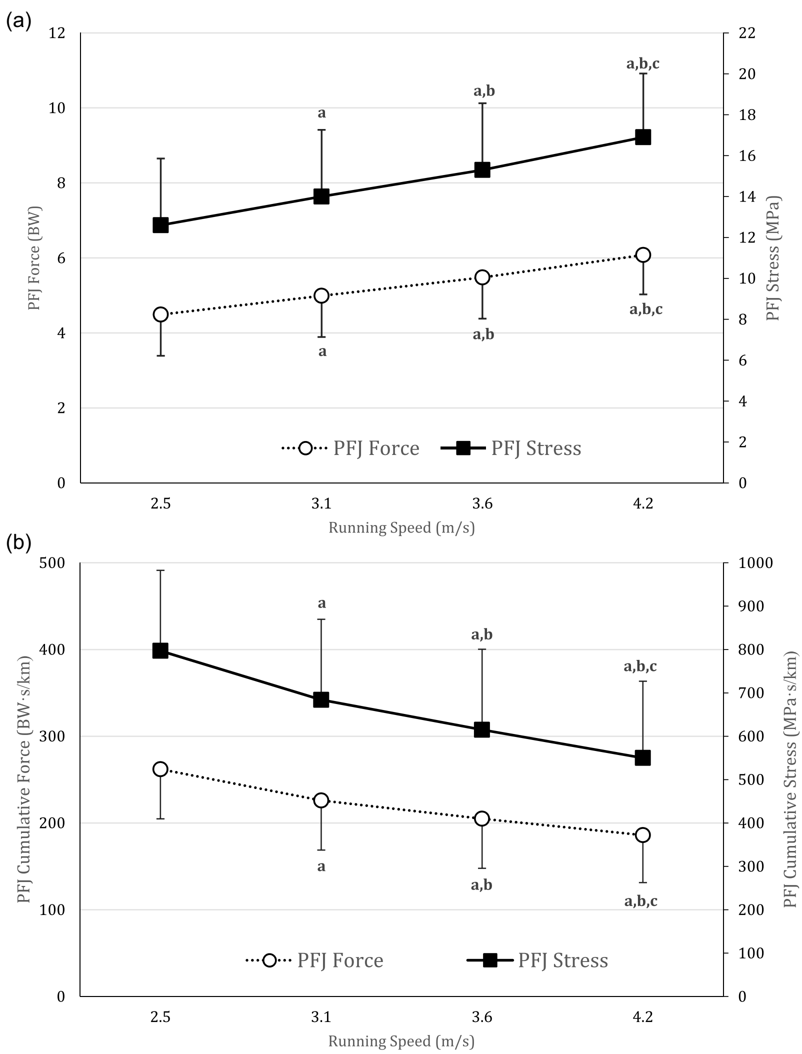

Figure 1. (a) Peak and (b) cumulative patellofemoral joint force and stress for each running speed. Abbreviations: PFJ, patellofemoral joint; BW, body weight. [a]Indicates significant difference from 2.5 m/s; [b]Indicates significant difference from 3.1 m/s; [c]Indicates significant difference from 3.6 m/s.

a 21.4–34.1% increase in PFJ stress (ES 0.82–1.32). There were significant differences between PFJ force and stress with each speed increment ($p < 0.001$) but with only small to moderate effect sizes between the two faster running speeds (3.6 and 4.2 m/s; ES 0.48–0.55). In contrast, estimated cumulative PFJ force ($p < 0.001$) and stress ($p < 0.001$) both significantly decreased with running speed (Figure 1(b)). The largest decrease in cumulative PFJ force and stress was observed between the two slowest speeds (2.5 and 3.1 m/s), with a 13.7–14.2% decrease (ES 0.61–0.63). Additional cumulative PFJ force and stress reductions were observed with each speed increment, albeit with only small effect sizes between each speed increase (ES 0.35–0.37; 9.3–10.6% decrease).

Spatiotemporal variables, kinematics, and kinetics across running speeds are presented in Table 3. Step rate, step length, peak GRF, IVLR, and peak knee flexion during stance significantly increased with each speed increment ($p < 0.001$). Knee flexion at initial contact was 1.0° ($p = 0.008$; ES 0.25) greater at 3.6 m/s and 1.6° ($p < 0.001$; ES 0.43) greater at 4.2 m/s compared to 2.5 m/s. Additionally, peak knee extensor moment significantly increased by 12.4% between 3.6 m/s ($p < 0.001$; ES 0.71) and 19.6% between 4.2 m/s ($p < 0.001$; ES 0.71) compared to 2.5 m/s. PFJ force and stress impulse did not significantly change between any running speed increments ($p \geq 0.25$), except for a small increase in PFJ force impulse between the slowest (2.5 m/s) and fastest (4.2 m/s) running speeds ($p = 0.02$; ES 0.26). In contrast, the number of steps required to complete 1 km of running significantly decreased with faster running speeds ($p <$ 0001; ES 0.53–0.71).

Table 3. Spatiotemporal variables, kinematics, and kinetics for each running speed (mean ± standard deviation).

	Running speed					Effect size (Cohen's *d*)		
	2.5 m/s	3.1 m/s	3.6 m/s	4.2 m/s	*p*	2.5 vs 3.1 m/s	2.5 vs 3.6 m/s	2.5 vs 4.2 m/s
Step rate, steps per min	170 ± 8	175 ± 9^{a}	182 ± 11^{a,b}	187 ± 10^{a,b,c}	<.001	0.53	1.24	1.91
Step length, m	0.89 ± 0.05	1.05 ± 0.06^{a}	1.20 ± 0.07^{a,b}	1.34 ± 0.07^{a,b,c}	<.001	2.64	4.99	7.46
Number of steps per 1 km	563 ± 30.1	476 ± 26.7^{a}	419 ± 24.7^{a,b}	374 ± 19.9^{a,b,c}	<.001	3.36	5.58	7.44
Peak GRF, BW	2.42 ± 0.19	2.56 ± 0.20^{a}	2.64 ± 0.21^{a,b}	2.69 ± 0.22^{a,b,c}	<.001	0.68	1.05	1.43
IVLR, BW/s	67.7 ± 14.7	83.6 ± 21.6^{a}	98.9 ± 26.4^{a,b}	126.0 ± 28.2^{a,b,c}	<.001	0.69	1.34	2.45
Peak knee flexion during stance, degrees	28.5 ± 7.0	30.7 ± 7.0^{a}	32.8 ± 7.2^{a,b}	34.7 ± 8.0^{a,b,c}	<.001	0.31	0.59	0.90
Knee flexion at IC, degrees	15.5 ± 3.6	16.0 ± 3.7	16.5 ± 4.27^{a}	17.2 4.5^{a,b}	<.001	0.12	0.25	0.43
Peak knee extensor moment (Nm/kg·m)	1.05 ± 0.21	1.12 ± 0.17^{a}	1.18 ± 0.18^{a,b}	1.27 ± 0.17^{a,b,c}	<.001	0.39	0.71	1.04
PFJ Force Impulse, BW.s	0.47 ± 0.13	0.48 ± 0.12	0.49 ± 0.14	0.52 ± 0.13^{a}	.08	0.07	0.18	0.26
PFJ Stress Impulse, MPa.s	1.42 ± 0.41	1.45 ± 0.41	1.48 ± 0.45	1.54 ± 0.44	.229	0.05	0.12	0.15

Abbreviations: BW, body weight; PFJ, patellofemoral joint; GRF, ground reaction force; IC, initial contact; VLR, vertical loading rate. Superscript indicates a significant difference with the following running speeds: ^{a}2.5 m/s; ^{b}3.1 m/s; ^{c}3.6 m/s.

Discussion and implications

This study investigated whether running speed influences peak and cumulative PFJ force and stress in recreational runners. The results indicate that increases in running speed had opposite effects on peak and cumulative PFJ kinetic outcomes. In line with our hypothesis, PFJ force and stress accumulated per 1 km of continuous running decreased despite the higher peak PFJ force and stress observed per step during faster running speeds. These factors should be carefully considered within the management and prevention of PFP.

The reduction in cumulative PFJ kinetics per km at faster speeds can be explained by the lower number of steps accumulated per km and the relatively unchanged PFJ force and stress impulse during faster running compared to slower running. This finding is consistent with previous research that found a 27% reduction in cumulative knee joint extensor moment impulse between 2.2 and 3.3 m/s (Petersen et al., 2015). In contrast, a previous study by Starbuck et al. reported that weighted cumulative PFJ force increased between 3.3 and 3.9 m/s, with no differences observed with further speed increments between 3.9 and 5.6 m/s (Starbuck et al., 2021). These results were observed in highly trained runners with a mean 10 km race time of 32.2 minutes compared to the mean time of 43.2 min for runners in this study. Notably, the largest decline in cumulative PFJ force and stress in the current study occurred between the two slowest running speeds (2.5 and 3.1 m/s), with smaller reductions occurring with each successive speed increase beyond 3.1 m/s. These collective findings across studies highlight a potential diminishing speed effect on cumulative PFJ measures at faster compared to slower running speeds. As a result, increasing speed to reduce cumulative PFJ force and stress per km is likely most important for runners changing from jogging speeds (~2.5 m/s) to slow running speeds (~3.1 m/s). These speeds are relevant to many recreational runners who represent a population of runners at high risk of PFP and may more accurately reflect their typical training speeds compared to faster speeds from previous studies. Additionally, large increases in running speed are less likely to be sustainable for typical training distances and more likely to cause detrimental effects of fatigue on running biomechanics compared to small increases in speed. Mean values for peak PFJ force and stress in the current study were in general agreement with previous literature that also accounted for co-contraction of the knee musculature (Chen & Powers, 2014; Willy et al., 2016). A recent meta-analysis by Hart et al. (2022) reported pooled PFJ forces of 5.2 times body weight in healthy runners across 27 studies with average running speeds similar to this study ranging from 2.3 m/s to 4.5 m/s (Hart et al., 2022). As running speed increased between 2.5 and 4.2 m/s, there was approximately a 35% increase in peak PFJ force and stress. There were also increases in knee extensor moment and peak knee flexion during stance between these two speeds contributing to the higher observed PFJ kinetics at faster running speeds. Previous studies have highlighted that runners with PFP typically ambulate at slower running speeds and with reduced knee flexion angles and knee extension moments (Bazett-Jones et al., 2023; Yang et al., 2022). It is unclear whether these biomechanical differences are compensatory strategies or if they occur before symptom onset (Bazett-Jones et al., 2023).

The most recent consensus statement on PFP (Collins et al., 2018) suggests a multimodal approach for runners for PFP, such as education, exercise, and running-specific interventions that reduce PFJ force (Esculier et al., 2020). International experts have advocated patient education as an important treatment component in runners with PFP and critical to successful management (Barton et al., 2015). However, no strong evidence from randomised controlled trials supports the benefits of education compared to control or wait-and-see approach when treating PFP (Willy et al., 2019). Typically, training advice consists of recommendations to avoid training loads that exceed the zone of tissues' capacity (Esculier et al., 2018). Based on the results from this study, coaches and clinicians wishing to minimise PFJ cumulative loading should consider prescribing set training distances with moderate running speeds of approximately 3.1 m/s rather than recommending continuous, slow jogging speeds (~2.5 m/s). An interval-based approach with running speeds at or above 3.1 m/s would also minimise accumulated PFJ load for a set training distance. Using this interval approach, the target training distance (i.e., 5 km) would need to be divided into shorter increments (e.g., 5 × 1 km) to account for the fatigue likely experienced when attempting to run long distances continuously at high speed. The optimisation of interval training approaches should be explored in future studies.

Overall training duration will be reduced when applying the higher running speed training approaches suggested by this study. This reduction relates to the set training distances being achieved in a shorter time compared to slower training speeds. Training duration will be reduced by a magnitude of speed1/speed2 where speed1 is the initial speed and speed2 is the increased speed. As a result, runners and coaches wishing to develop a cumulative PFJ load management programme based on duration rather than distance should adjust target training durations by this factor when increasing training speed. Additionally, while these approaches reduce total accumulated force and stress, the peak PFJ force and stress for each step will be increased. These increased peak forces and stresses are expected to be within a physiologically tolerable range; however, if PFJ symptoms are aggravated, then proposed increases to running speed could be combined with other supplementary strategies shown to reduce PFJ peak force and stress (e.g., recommendations to implement gait retraining strategies, such as increasing step rate (Doyle et al., 2022) or use minimalist footwear (Bonacci et al., 2014; Zhang et al., 2022)). These suggestions to manipulate training speeds may be relevant in preventing and managing overuse PFJ conditions and should be explored further in prospective cohort and intervention studies.

It is important to observe some limitations in this study. Modelling approaches to assess PFJ force and stress have been extensively discussed in the literature (Nunes et al., 2018) and may impact the findings of this study. Our PFJ model was a simplified planar model that does not consider individual three-dimensional patella kinematics and geometry. These individual differences may affect PFJ contact area and stress. Tissue properties were not accounted for in our modelling approach and should be considered when interpreting our findings. Increasing running speed reduced the load cycles and cumulative load per km, but the magnitude of each load cycle was greater, which may warrant greater weighting when attempting to optimise knee tissue health, particularly over long distances. A further limitation is that all running was performed indoors on an instrumented treadmill on a flat gradient, and caution should be undertaken when translating findings to overground or field-based running. While estimations of PFJ force and stress are reported to be similar across treadmill and overground running

(Willy et al., 2016), the effects of surface and gradient changes at various running speeds may also influence the relationships observed in this study. A final limitation is that cumulative PFJ measures were determined using the total impulse of the respective PFJ force and stress curves, which assume equal weighting applied across the gait cycle. A weighted-cumulative approach that factors in the magnitude of force and the fatigue behaviour of the PFJ could be considered in future research, although the authors are not aware of a current method for this in the literature. Inferences to injury risk based on cumulative load should therefore be interpreted cautiously.

Conclusion

Peak PFJ force and stress significantly increased with faster running speeds, while cumulative PFJ force and stress decreased. The findings suggest that while faster running increases the magnitude of PFJ force and stress peaks per step; it also results in fewer steps, no differences in PFJ force and stress impulse, and less total accumulated force and stress.

Coaches and clinicians working with recreational runners should consider prescribing moderate running speeds of approximately 3.1 m/s with reduced training duration or an interval-based approach instead of continuous, slow jogging speeds (~2.5 m/s) to manage cumulative measures of PFJ force and stress. This information may be relevant in preventing and managing overuse PFJ conditions and should be further investigated through prospective cohort and intervention studies.

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Data availability statement

Data are available on request from the authors. The data supporting the study findings are available from the corresponding author, upon reasonable request, by emailing Eoin Doyle (eoin.doyle@mq.edu.au).

References

Andersen, J. (2019). *The state of running*. https://runrepeat.com/state-of-running

Barton, C. J., Lack, S., Hemmings, S., Tufail, S., & Morrissey, D. (2015). The 'best practice guide to conservative management of patellofemoral pain': Incorporating level 1 evidence with expert clinical reasoning. *British Journal of Sports Medicine*, *49*(14), 923–934. https://doi.org/10.1136/bjsports-2014-093637

Bazett-Jones, D. M., Neal, B. S., Legg, C., Hart, H. F., Collins, N. J., & Barton, C. J. (2023). Kinematic and kinetic gait characteristics in people with patellofemoral pain: A systematic review and meta-analysis. *Sports Medicine*, *53*(2), 519–547. https://doi.org/10.1007/s40279-022-01781-1

Bell, A. L., Pedersen, D. R., & Brand, R. A. (1990). A comparison of the accuracy of several hip center location prediction methods. *Journal of Biomechanics*, *23*(6), 617–621. https://doi.org/10.1016/0021-9290(90)90054-7

Bonacci, J., Hall, M., Fox, A., Saunders, N., Shipsides, T., & Vicenzino, B. (2018). The influence of cadence and shoes on patellofemoral joint kinetics in runners with patellofemoral pain. *Journal of Science & Medicine in Sport*, *21*(6), 574–578. https://doi.org/10.1016/j.jsams.2017.09.593

Bonacci, J., Vicenzino, B., Spratford, W., & Collins, P. (2014). Take your shoes off to reduce patellofemoral joint stress during running. *British Journal of Sports Medicine*, *48*(6), 425–428. https://doi.org/10.1136/bjsports-2013-092160

Buist, I., Bredeweg, S. W., Bessem, B., van Mechelen, W., Lemmink, K. A., & Diercks, R. L. (2010). Incidence and risk factors of running-related injuries during preparation for a 4-mile recreational running event. *British Journal of Sports Medicine*, *44*(8), 598–604. https://doi.org/10.1136/bjsm.2007.044677

Cappozzo, A., Catani, F., Croce, U. D., & Leardini, A. (1995). Position and orientation in space of bones during movement: Anatomical frame definition and determination. *Clinical Biomechanics*, *10*(4), 171–178. https://doi.org/10.1016/0268-0033(95)91394-T

Chen, Y. J., & Powers, C. M. (2014). Comparison of three-dimensional patellofemoral joint reaction forces in persons with and without patellofemoral pain. *Journal of Applied Biomechanics*, *30*(4), 493–500. https://doi.org/10.1123/jab.2011-0250

Chin, A., Elliott, B., Alderson, J., Lloyd, D., & Foster, D. (2009). The off-break and "doosra": Kinematic variations of elite and sub-elite bowlers in creating ball spin in cricket bowling. *Sports Biomechanics*, *8*(3), 187–198. https://doi.org/10.1080/14763140903229476

Cohen, J. (2013). *Statistical power analysis for the behavioral sciences*.

Collins, N. J., Barton, C. J., van Middelkoop, M., Callaghan, M. J., Rathleff, M. S., Vicenzino, B. T., Davis, I. S., Powers, C. M., Macri, E. M., Hart, H. F., de Oliveira Silva, D., & Crossley, K. M. (2018). Consensus statement on exercise therapy and physical interventions (orthoses, taping and manual therapy) to treat patellofemoral pain: Recommendations from the 5th international patellofemoral pain research retreat, Gold Coast, Australia, 2017. *British Journal of Sports Medicine*, *52*(18), 1170–1178. https://doi.org/10.1136/bjsports-2018-099397

Crossley, K. M., van Middelkoop, M., Callaghan, M. J., Collins, N. J., Rathleff, M. S., & Barton, C. J. (2016). Patellofemoral pain consensus statement from the 4th international patellofemoral pain research retreat, Manchester. Part 2: Recommended physical interventions (exercise, taping, bracing, foot orthoses and combined interventions). *British Journal of Sports Medicine*, *50*(14), 844–852. https://doi.org/10.1136/bjsports-2016-096268

DeVita, P., & Hortobagyi, T. (2001). Functional knee brace alters predicted knee muscle and joint forces in people with ACL reconstruction during walking. *Journal of Applied Biomechanics*, *17*(4), 297–311. https://doi.org/10.1123/jab.17.4.297

Doyle, E., Doyle, T. L. A., Bonacci, J., & Fuller, J. T. (2022). The effectiveness of gait retraining on running kinematics, kinetics, performance, pain, and injury in distance runners: A systematic review with meta-analysis. *Journal of Orthopaedic & Sports Physical Therapy*, *52*(4), 192–A195. https://doi.org/10.2519/jospt.2022.10585

Esculier, J. F., Bouyer, L. J., Dubois, B., Fremont, P., Moore, L., McFadyen, B., & Roy, J.-S. (2018). Is combining gait retraining or an exercise programme with education better than education alone in treating runners with patellofemoral pain? A randomised clinical trial. *British Journal of Sports Medicine*, *52*(10), 659–666. https://doi.org/10.1136/bjsports-2016-096988

Esculier, J. F., Maggs, K., Maggs, E., & Dubois, B. (2020). A contemporary approach to patellofemoral pain in runners. *Journal of Athletic Training*, *55*(12), 1206–1214. https://doi.org/10.4085/1062-6050-0535.19

Francis, P., Whatman, C., Sheerin, K., Hume, P., & Johnson, M. I. (2019). The proportion of lower limb running injuries by gender, anatomical location and specific pathology: A systematic review. *Journal of Sports Science & Medicine*, *18*(1), 21–31.

Futrell, E. E., Jamison, S. T., Tenforde, A. S., & Davis, I. S. (2018). Relationships between habitual cadence, footstrike, and vertical load rates in runners. *Medicine & Science in Sports & Exercise*, *50*(9), 1837–1841. https://doi.org/10.1249/MSS.0000000000001629

Hart, H. F., Patterson, B. E., Crossley, K. M., Culvenor, A. G., Khan, M. C. M., King, M. G., & Sritharan, P. (2022). May the force be with you: Understanding how patellofemoral joint reaction force compares across different activities and physical interventions—A systematic review and meta-analysis. *British Journal of Sports Medicine*, *56*(9), 521–530. https://doi.org/10.1136/bjsports-2021-104686

Heino Brechter, J., & Powers, C. M. (2002). Patellofemoral stress during walking in persons with and without patellofemoral pain. *Medicine & Science in Sports & Exercise*, *34*(10), 1582–1593. https://doi.org/10.1097/00005768-200210000-00009

Hopkins, W. G., Schabort, E. J., & Hawley, J. A. (2001). Reliability of power in physical performance tests. *Sports Medicine*, *31*(3), 211–234. https://doi.org/10.2165/00007256-200131030-00005

Lankhorst, N. E., van Middelkoop, M., Crossley, K. M., Bierma-Zeinstra, S. M. A., Oei, E. H. G., Vicenzino, B., & Collins, N J. (2016). Factors that predict a poor outcome 5–8 years after the diagnosis of patellofemoral pain: A multicentre observational analysis. *British Journal of Sports Medicine*, *50*(14), 881–886. https://doi.org/10.1136/bjsports-2015-094664

Mai, P., & Willwacher, S. (2019). Effects of low-pass filter combinations on lower extremity joint moments in distance running. *Journal of Biomechanics*, *95*, 109311. https://doi.org/10.1016/j.jbiomech.2019.08.005

Napier, C., Willy, R. W., Hannigan, B. C., McCann, R., & Menon, C. (2021). The effect of footwear, running speed, and location on the validity of two commercially available inertial measurement units during running. *Frontiers in Sports and Active Living*, *3*, 643385. https://doi.org/10.3389/fspor.2021.643385

Nielsen, R. O., Nohr, E. A., Rasmussen, S., & Sorensen, H. (2013). Classifying running-related injuries based upon etiology, with emphasis on volume and pace. *International Journal of Sports Physical Therapy*, *8*(2), 172–179.

Nimon, G., Murray, D., Sandow, M., & Goodfellow, J. (1998). Natural history of anterior knee pain: A 14- to 20-year follow-up of nonoperative management. *Journal of Pediatric Orthopaedics*, *18*(1), 118–122. https://doi.org/10.1097/01241398-199801000-00021

Nunes, G. S., Scattone Silva, R., Dos Santos, A. F., Fernandes, R. A. S., Serrao, F. V., & de Noronha, M. (2018). Methods to assess patellofemoral joint stress: A systematic review. *Gait & Posture*, *61*, 188–196. https://doi.org/10.1016/j.gaitpost.2017.12.018

Oliveira, A. S., & Pirscoveanu, C. I. (2021). Implications of sample size and acquired number of steps to investigate running biomechanics. *Scientific Reports*, *11*(1), 3083. https://doi.org/10.1038/s41598-021-82876-z

Petersen, J., Sorensen, H., & Nielsen, R. O. (2015). Cumulative loads increase at the knee joint with slow-speed running compared to faster running: A biomechanical study. *Journal of Orthopaedic & Sports Physical Therapy*, *45*(4), 316–322. https://doi.org/10.2519/jospt.2015.5469

Powers, C. M., Bolgla, L. A., Callaghan, M. J., Collins, N., & Sheehan, F. T. (2012). Patellofemoral pain: Proximal, distal, and local factors, 2nd international research retreat. *Journal of Orthopaedic & Sports Physical Therapy*, *42*(6), A1–54. https://doi.org/10.2519/jospt.2012.0301

Rathleff, M. S., Rathleff, C. R., Olesen, J. L., Rasmussen, S., & Roos, E. M. (2016). Is knee pain during adolescence a self-limiting condition?: Prognosis of patellofemoral pain and other types of knee pain. *The American Journal of Sports Medicine*, *44*(5), 1165–1171. https://doi.org/10.1177/0363546515622456

Robertson, D. G., & Dowling, J. J. (2003). Design and responses of Butterworth and critically damped digital filters. *Journal of Electromyography and Kinesiology*, *13*(6), 569–573. https://doi.org/10.1016/S1050-6411(03)00080-4

Smith, B. E., Moffatt, F., Hendrick, P., Bateman, M., Rathleff, M. S., Selfe, J., Smith, T. O., & Logan, P. (2018). The experience of living with patellofemoral pain—loss, confusion and fear-avoidance: A UK qualitative study. *British Medical Journal Open*, *8*(1), e018624. https://doi.org/10.1136/bmjopen-2017-018624

Smith, B. E., Selfe, J., Thacker, D., Hendrick, P., Bateman, M., Moffatt, F., Rathleff, M. S., Smith, T. O., & Logan, P. (2018). Incidence and prevalence of patellofemoral pain: A systematic review and meta-analysis. *PLoS One*, *13*(1), e0190892. https://doi.org/10.1371/journal.pone.0190892

Starbuck, C., Bramah, C., Herrington, L., & Jones, R. (2021). The effect of speed on achilles tendon forces and patellofemoral joint stresses in high-performing endurance runners. *Scandinavian Journal of Medicine & Science in Sports*, *31*(8), 1657–1665. https://doi.org/10.1111/sms.13972

Tanner, R. K., & Gore, C. J. (2013). *Physiological tests for elite athletes* (2nd ed.). Human Kinetics.

van Eijden, T. M., Kouwenhoven, E., Verburg, J., & Weijs, W. A. (1986). A mathematical model of the patellofemoral joint. *Journal of Biomechanics*, *19*(3), 219–229. https://doi.org/10.1016/0021-9290(86)90154-5

Willy, R. W., Halsey, L., Hayek, A., Johnson, H., & Willson, J. D. (2016). Patellofemoral joint and achilles tendon loads during overground and treadmill running. *Journal of Orthopaedic & Sports Physical Therapy*, *46*(8), 664–672. https://doi.org/10.2519/jospt.2016.6494

Willy, R. W., Hoglund, L. T., Barton, C. J., Bolgla, L. A., Scalzitti, D. A., Logerstedt, D. S., Lynch, A. D., Snyder Mackler, L., & McDonough, C. M. (2019). Patellofemoral pain. *Journal of Orthopaedic & Sports Physical Therapy*, *49*(9), CPG1–CPG95. https://doi.org/10.2519/jospt.2019.0302

Woltring, H. J. (1985). On optimal smoothing and derivative estimation from noisy displacement data in biomechanics. *Human Movement Science*, *4*(3), 229–245. https://doi.org/10.1016/0167-9457(85)90004-1

Yang, C., Best, T. M., Liu, H., & Yu, B. (2022). Knee biomechanical factors associated with patellofemoral pain in recreational runners. *The Knee*, *35*, 87–97. https://doi.org/10.1016/j.knee.2022.02.007

Zhang, M., Zhou, X., Zhang, L., Liu, H., & Yu, B. (2022). The effect of heel-to-toe drop of running shoes on patellofemoral joint stress during running. *Gait & Posture*, *93*, 230–234. https://doi.org/10.1016/j.gaitpost.2022.02.008